

AMCS2034

Tutorial 4: Time and Space Complexity

- Q1. Explain time complexity and space complexity.
- Q2. Describe the steps in doing empirical analysis.
- Q3. Why is performing comparative timing of programs difficult?
- Q4. How does code tuning affect the running times of two programs with equal growth rates?
- Q5. Explain simulation in terms of algorithm analysis.
- Q6. State any **FOUR (4)** primitive operations that can be counted in performing an algorithm analysis.
- Q7. Explain the space/time tradeoff principle in algorithm design.
- Q8. Describe the role of *lookup tables* in relation to the space/time tradeoff principle.
- Q9. Discuss the *disk-based* space/time tradeoff principle.